

Systems of Linear Equations

Section 1.1

Why Do We Study Systems of Equations?

Many real-world problems involve finding unknown values subject to multiple constraints.

Examples:

- allocating budgets across competing needs
- managing traffic flow at intersections
- planning production under material, time, and storage limits

Linear Equations

A **linear equation** in the variables $x_1, \dots, x_n$ is an equation that can be written in the form

$$a_1x_1 + a_2x_2 + \dots + a_nx_n = b$$

where b and the **coefficients** $a_1, \dots, a_n$ are real or complex numbers.

Systems of Linear Equations

Definition

A **system of linear equations** (or **linear system**) is a collection of one or more linear equations involving the same variables.

Solutions of a Linear System

Definition

A **solution** of the system is a list of numbers, or n -tuple, $(s_1, s_2, \dots, s_n)$ that makes each equation a true statement when the values of $s_1, s_2, \dots, s_n$ are substituted for $x_1, \dots, x_n$, respectively.

Solutions of a Linear System

Definition

The set of all possible solutions to a system of linear equations is called the **solution set** of the linear system.

* Two linear systems are said to be **equivalent** if they have the same solution set.

Solutions of a Linear System

Example:

Nature of Solutions

A system of linear equations has either:

1. no solution
2. exactly one solution
3. infinitely many solutions

A linear system is said to be **consistent** if it has at least one solution; otherwise, the linear system is **inconsistent**.

Matrix Notation

The essential information from a linear system can be stored compactly as a rectangular array of numbers called a **matrix**.

An $m \times n$ **matrix** is a matrix with m rows and n columns.

Solving a Linear System

To solve a linear system, we reduce the given system to an equivalent system that is easier to solve using the following basic operations:

1. Replace one equation by the sum of itself and a multiple of another equation.
2. Interchange any two equations.
3. Multiply all terms in an equation by a nonzero constant.

Solving a Linear System

Example:

Elementary Row Operations

(Corresponding to basic operations applied to a linear system)

1. **Replacement:** Replace one row by the sum of itself and a multiple of another row.
2. **Interchange:** Interchange two rows.
3. **Scaling:** Multiply all entries in a row by a nonzero constant.

Notes

- Two matrices are said to be **row equivalent** if there exists a sequence of elementary row operations which transforms one matrix into the other.
- If the augmented matrices of two linear systems are row equivalent, then the two systems have the same solution set.

Existence and Uniqueness Questions

Two Fundamental Questions about a Linear System

1. Is the system consistent; i.e., does at least one solution *exist*?
2. If a solution exists, is it the *only* one; i.e., is the solution *unique*?